

RES.6-012 Intro to Probability - Lecture 14, 15
Textbook Notes - Sections 8.1 & 8.2.

◦ Section 8.1 Bayesian Inference & the Posterior Distribution.

• The Bayesian Framework:

The Bayesian approach models an unknown parameter θ as a realisation of a random variable Θ with a known prior distribution $p_{\Theta}(\theta)$ or $f_{\Theta}(\theta)$

The setup has 3 components:

[a] Prior distribution p_{Θ} or f_{Θ} encoding initial beliefs about Θ .

[b] Observation model $p_{X|\Theta}$ or $f_{X|\Theta}$ specifying how the data $X = (X_1, \dots, X_n)$ depend on Θ

[c] Posterior distribution $p_{\Theta|X}(\theta|x)$ or $f_{\Theta|X}(\theta|x)$ (after observing value x of X) is computed via Bayes' rule.

↳ Posterior encapsulates everything that data knows about Θ .

→ The posterior is the complete answer to the Bayesian inference problem.
It can then be used to:

- ↳ derive point estimates
- ↳ compute error probabilities
- ↳ answer further inference questions.

• Four Versions of Bayes Rule

All four are identical $\rightarrow$ replace PMF with PDF

$\rightarrow$ replace sum with integral

① θ discrete, X discrete

$$P_{\theta|X}(\theta|x) = \frac{P_{\theta}(\theta) \cdot p_{x|\theta}(x|\theta)}{\sum_{\theta'} P_{\theta}(\theta') \cdot p_{x|\theta}(x|\theta')}$$

② θ discrete, X continuous

$$P_{\theta|X}(\theta|x) = \frac{P_{\theta}(\theta) \cdot f_{x|\theta}(x|\theta)}{\sum_{\theta'} P_{\theta}(\theta') \cdot f_{x|\theta}(x|\theta')}$$

③ θ continuous, X discrete

$$f_{\theta|x}(\theta|x) = \frac{f_{\theta}(\theta) \cdot p_{x|\theta}(x|\theta)}{\int f_{\theta}(\theta') \cdot p_{x|\theta}(x|\theta') d\theta'}$$

④ θ continuous, X continuous

$$f_{\theta|x}(\theta|x) = \frac{f_{\theta}(\theta) \cdot f_{x|\theta}(x|\theta)}{\int f_{\theta}(\theta') \cdot f_{x|\theta}(x|\theta') d\theta'}$$

$\hookrightarrow$ When θ is multidimensional, the sums/integrals become multidimensional.

Key !! $\rightarrow$ Denominator does not depend on θ

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$\hookrightarrow$ normalising constant ensures posterior integrates/sums to 1.

⑤ For Maximisation (MAP), denominator can be ignored entirely.

• Examples

↳ Romeo & Juliet

Juliet late on any date by an amount X uniformly distributed on $[0, \theta]$.

Parameter θ is modelled as $\theta \sim \text{Uniform}[0, 1]$

① Prior: $f_{\theta}(\theta) = 1$ for $0 \leq \theta \leq 1$.

② Observation model: $f_{x|\theta}(x|\theta) = \frac{1}{\theta}$ for $0 \leq x \leq \theta$.

③ Compute Posterior: After one date where Juliet is late by x , the posterior is (for $x \leq \theta \leq 1$):

$$f_{\theta|x}(\theta|x) = \frac{(1/\theta)}{\int_x^1 1/\theta' d\theta'} = \frac{1}{\theta \cdot |\log x|}$$

For n dates with observations $x_1, \dots, x_n$ (conditionally indep. uniform $[0, \theta]$) the likelihood is $1/\theta^n$ for $\theta \geq \bar{x} = \max\{x_1, \dots, x_n\}$, giving posterior:

$$f_{\theta|x}(\theta|x) = \frac{c(\bar{x})}{\theta^n}, \quad \bar{x} \leq \theta \leq 1$$

where $c(\bar{x})$ is a normalising constant.

↳ Normal Prior with Normal Observations

Observations $X_i = \theta + W_i$ where $\theta \sim N(x_0, \sigma_0^2)$ and $W_i \sim N(0, \sigma_i^2)$ are all independent.

Result: The posterior $f_{\theta|x}(\theta|x)$ is Normal with mean and variance:

$$m = E[\theta | X=x] = \frac{\sum_{i=0}^n x_i / \sigma_i^2}{\sum_{i=0}^n 1/\sigma_i^2}, \quad v = \text{var}(\theta | X=x) = \frac{1}{\sum_{i=0}^n 1/\sigma_i^2}$$

The posterior mean is a precision-weighted average of the prior mean x_0 and the observations $x_1, \dots, x_n$. Higher-precision (lower-variance) observations receive more weight.

Special case: $(\sigma_0^2 = \sigma_1^2 = \dots = \sigma_n^2 = \sigma^2)$

- Posterior mean $m = (x_0 + x_1 + \dots + x_n) / (n+1)$ and posterior variance $v = \frac{\sigma^2}{(n+1)}$.
- The prior mean x_0 acts as an additional observation with equal weight.
- The posterior standard deviation shrinks at rate $\frac{1}{\sqrt{n}}$

Conjugate Prior Property → the posterior is Normal (same as prior).
↳ enables recursive updating:
when a new observation x_{n+1} arrives,
current posterior acts as new prior
↳ update formula applied directly without re-solving from scratch.

• Multiparameter Problems

The framework extends identically to $\Theta = (\Theta_1, \dots, \Theta_k)$.

The prior $f_{\Theta}(\Theta_1, \dots, \Theta_k)$ is now a joint distribution. ~~B~~

↳ Bayes' rule applied with multidimensional integrals in the denominator.

• Section 8.2 Point Estimation, Hypothesis Testing, & the MAP Rule

• Point Estimation Setup

The posterior distribution is the complete answer to Bayesian Inference, however a single reference number is often desired (a point estimate).

Estimator $\rightarrow$ A RV $\hat{\theta} = g(X)$

$\rightarrow$ a function of the observations, before the data is seen.

Estimation $\rightarrow$ The numerical value $\hat{\theta} = g(x)$

$\rightarrow$ The output once the data x is observed.

Two principle estimators:

(a) MAP (Maximum a Posteriori Probability) estimator

$\rightarrow$ choose $\hat{\theta}$ to maximise the posterior $p_{\theta|x}(\theta|x)$ (discrete)
or $f_{\theta|x}(\theta|x)$ (continuous)

(b) Conditional Expectation (LMS) estimator

$\rightarrow$ choose $\hat{\theta} = E[\theta|X=x]$

$\rightarrow$ minimises the mean squared error

$\Rightarrow$ When the posterior is symmetric and unimodal (e.g. Normal), the two estimators ~~coincide~~ coincide. For skewed posteriors, they differ.

• The MAP Rule

Given observation x , select $\hat{\theta}$ that maximises the posterior over θ :

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} f_{\theta|x}(\theta|x) \quad (\theta \text{ continuous})$$

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p_{\theta|x}(\theta|x) \quad (\theta \text{ discrete})$$

Shortcut: Since Bayes' rule denominator does not depend on θ , maximising the posterior is equivalent to maximising the numerator.

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} f_{\theta}(\theta) \cdot f_{x|\theta}(x|\theta)$$

(with appropriate PMF/PDF replacements depending on the type of θ and X).

Optimality: When θ takes finitely many values, the MAP rule minimises the probability of incorrect decision, both conditional probability of error given any observations x , and the overall (unconditional) probability of error over all decision rules.

• MAP Estimation Examples

→ Normal posterior

Posterior is Normal with mean m and variance v .

A ~~an~~ Normal PDF is maximised at its mean, so $\theta_{\text{MAP}} = m$.

↳ equals conditional expectation $E[\theta | X=x]$.

→ Romeo & Juliet

The posterior $f_{\theta|X}(\theta|x) = \frac{1}{(\theta \cdot |\log x|)}$ is decreasing in θ over $[x, 1]$. The MAP estimate is $\hat{\theta} = x$ (max at left endpoint).

Conditional Expectation estimate:

$$E[\theta | X=x] = \frac{1-x}{|\log x|}$$

↳ which is larger than x for all $x \in (0, 1)$.

The MAP estimate is "optimistic" (predicts future lateness equals observed lateness)

↳ The Conditional expectation is more conservative.

• Hypothesis Testing

When Θ takes one of m discrete values $\theta_1, \dots, \theta_m$, the inference problem becomes: hypothesis testing.

The event $\{\Theta = \theta_i\}$ is the i -th hypothesis H_i .

The MAP rule selects the hypothesis with the largest posterior probability:

→ Select H_i for which $P(\Theta = \theta_i | X = x)$ is largest.

Equivalently, select H_i for which $p_{\Theta}(\theta_i) \cdot p_{X|\Theta}(x|\theta_i)$ is largest.

Optimality: The MAP rule minimises the overall probability of error over all possible decision rules.

The probability of correct decision for observation x is $P(\Theta = \hat{\Theta}_{\text{MAP}}(x) | X = x)$

The overall probability of error is:

$$P(\text{error}) = 1 - P(\Theta = \hat{\Theta}_{\text{MAP}}(X))$$

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Textbook Notes - Sections 8.3, 8.4

• Architecture:

$$\underbrace{E[\theta | x]}_{\text{8.3 LMS estimator}} \xrightarrow{\text{minimise}} \underbrace{E[(\theta - \hat{\theta})^2]}_{\text{MSE}} \xrightarrow{\text{restrict to linear}} \underbrace{\hat{\theta}_L = E[\theta] + \frac{\text{cov}(\theta, x)}{\text{var}(x)}(x - E[x])}_{\text{8.4 Linear LMS Estimator}}$$

• Section 8.3 Bayesian Least Mean Squares (LMS) Estimation

• The LMS Principle

Without any observation, the estimator $\hat{\theta}$ minimising $E[(\theta - \hat{\theta})^2]$ is $\hat{\theta} = E[\theta]$
→ the prior mean.

It follows from $E[(\theta - \hat{\theta})^2] = \text{var}(\theta) + (E[\theta] - \hat{\theta})^2$, minimised when $\hat{\theta} = E[\theta]$.

Given observation $X = x$, the same argumented conditionally:
 $\hat{\theta} = E[\theta | X = x]$

estimator

The LMS estimator $\hat{\theta} = E[\theta | X]$ minimises the MSE over all possible estimators $g(X)$:

$$E[(\theta - E[\theta | X])^2] \leq E[(\theta - g(X))^2] \text{ for all } g.$$

• Properties of the LMS Estimation Error

Let $\hat{\theta} = E[\theta | X]$ and $\tilde{\theta} = \hat{\theta} - \theta$

• Zero mean (unbiased):

$$\hookrightarrow E[\tilde{\theta} | X = x] = 0 \text{ for all } x, \quad E[\tilde{\theta}] = 0$$

• Uncorrelated with the estimate

$$\text{cov}(\hat{\theta}, \tilde{\theta}) = 0$$

• Variance Decomposition

$$\text{var}(\theta) = \text{var}(\hat{\theta}) + \text{var}(\tilde{\theta})$$

• The total variability in θ splits into variability explained by

$\hookrightarrow X$ (through $\hat{\theta}$)

$\hookrightarrow$ residual uncertainty (through $\tilde{\theta}$)

• $\hookrightarrow$ Same properties as Section 4.3.

• Comparing MAP and LMS

MAP: maximises the posterior $f_{\theta|X}(\theta|x)$ - a mode.

LMS: takes the posterior mean $E[\theta | X = x]$ - minimise MSE.

When the posterior is symmetric and unimodal (e.g. Normal), they coincide.

Otherwise, LMS has uniformly smaller MSE.

Coin Example p. 435

With uniform prior and k heads in n tosses. ($\text{Beta}(k+1, n-k+1)$ posterior)

• MAP: $\hat{\theta} = k/n$

• LMS: $E[\theta | X = k] = (k+1)/(n+2)$

$\hookrightarrow$ LMS shrinks toward $1/2$; two estimates converge for large n .

• Multiple Observations / Parameters

For $X = (X_1, \dots, X_n)$, the LMS estimator is $E[\Theta | X_1, \dots, X_n]$
↳ same principle applied to the joint posterior

For multiple parameters $\Theta_1, \dots, \Theta_m$ the joint MSE criterion decouples into m independent LMS problems, one per parameter.

In practise $\rightarrow$ LMS computation requires (difficult!)
 $\rightarrow$ full joint PDF
 $\rightarrow$ may produce non-linear function of the observations.
↳ $f_{X|\Theta}(x|\theta) = \prod_{i=1}^n f_{X_i|\Theta}(x_i|\theta)$

• Section 8.4 Bayesian Linear LMS Estimation

• Setup

Restrict to estimators of the form $\hat{\Theta} = aX + b$.
Choose a and b to minimise $E[(\Theta - aX - b)^2]$

Step 1 $\rightarrow$ Optimal b given a : minimising over b gives $b = E[\Theta] - aE[X]$

Step 2 $\rightarrow$ Optimal a : substituting and minimising over a
$$a = \frac{\text{cov}(\Theta, X)}{\text{var}(X)}$$

• Linear LMS Formula (Single Observation)

$$\hat{\theta}_L = E[\theta] + \frac{\text{cov}(\theta, X)}{\text{var}(X)} \cdot (X - E[X]) = E[\theta] + \rho \cdot \frac{\sigma_\theta}{\sigma_X} (X - E[X])$$

↳ where $\rho = \text{cov}(\theta, X) / (\sigma_\theta \cdot \sigma_X)$ is the correlation coefficient.

Resulting MSE:

$$E[(\theta - \hat{\theta}_L)^2] = (1 - \rho^2) \cdot \sigma_\theta^2$$

- When $|\rho| \approx 1$ (highly correlated), the error is small.
- When $\rho = 0$ (uncorrelated), the observation is useless and the estimator reduces to $E[\theta]$.

Interpretation $\rightarrow$ the estimator starts from the prior mean and adjusts proportionally to how much X deviates from its own mean, scaled by the covariance.

• When LMS and Linear LMS Coincide:

If LMS estimator $E[\theta|X]$ happens to be linear in X , then it is also the linear LMS estimator.

↳ Happens in Normal model if X_n & θ are jointly Normal.

• Multiple Observations Formula

For observations $X_i = \Theta + W_i$ where $\Theta, W_1, \dots, W_n$ are uncorrelated with $E[W_i] = 0$ and $\text{var}(W_i) = \sigma_i^2$:

$$\hat{\Theta}_L = \frac{\mu/\sigma_0^2 + \sum_{i=1}^n \frac{X_i}{\sigma_i^2}}{\sum_{i=0}^n \frac{1}{\sigma_i^2}}, \quad \text{var}(\hat{\Theta}_L) = \frac{1}{\sum_{i=0}^n \frac{1}{\sigma_i^2}}$$

where $\mu = E[\Theta]$ and $\sigma_0^2 = \text{var}(\Theta)$

↳ Same precision-weighted average from the Normal-Normal conjugate update in Section 8.1.

Prior acts as an additional observation with precision $\frac{1}{\sigma_0^2}$

• Summary

Estimator	Criterion	Requires	MSE
MAP	maximise posterior	full posterior	not minimised
LMS	minimise MSE	full posterior	minimum possible
Linear LMS	minimise MSE over linear $\hat{\Theta} = aX + b$	means, variances, covariances only	$(1 - \rho^2) \cdot \sigma_\Theta^2$

↳ Under Normality $\rightarrow$ all three coincide.